

Hurrying Data Along

The society of the future will depend on fast networks more than ever before. Here is a look at that future

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Communication depends on networks. Period. While networks can mean many things in many contexts, it's the physical networks, the wires and the waves that enable it all. As the technological revolution proceeds at breakneck speed, more and more digital data is created than ever before. And this data isn't created by huge corporations, universities and governments—each one of us, when we send an SMS, click a picture or shoot off an e-mail, create data which will invariably be added to the repository of global digital data. In fact, IDC—a research body—estimates that 70 per cent of the 988 exabytes of data (one exabyte is equal to 50,000 hours of DVD quality video)—projected to be the global digital output by 2010—will be created by individuals.

Even by conservative estimates, that number is huge. The thing to note is that most of that data is not going to be static. It will exist in the Cloud—a term recently much in use and which roughly means the Internet. Everyday, as

humans upload more and more data from their gazillion devices, the networks that support these data transfers will need to get more robust, both in quantity and quality.

Inside The Chip

Before talking about megascales, let's focus a bit on nanoscales. Inside the processor, we have been seeing an increased concentration of cores. The latest processors have four cores, though manufacturers like Intel and AMD have released roadmaps for octal core processors by 2009. And then the stage will be set for 12, 15, 25....

With all these cores in a tiny sliver of silicon, networking and data transfers between cores in the die will need to have a quantum change. Unlike today's processor architecture where running different applications means that the processor switches rapidly between them, future designs with say, 100 cores will have separate cores devoted to individual processes. This means that encryption would get the attention of one core, video encoding would be the domain of another one, gaming will be handled by the third...you get the drift. In such a scenario,

current bus transfers data rates simply can't keep up. If we have to go with current architecture, that would mean that most of the cores would be starved of data, and instead of speeding up individual applications, they would slow down.

To get around the inter-chip networking bottlenecks, companies like Intel and Rambus have been working on prototypes of newer architectural models. Rambus has a program called Terabyte Bandwidth Initiative (TBI), while Intel's programme is called Terascale Computing Research Program. Last February, Intel has released an 80-core chip that delivers supercomputer-grade speeds on desktop computers. The fingernail-sized Teraflops Research Chip's performance metric is 1.81 Teraflops (trillion floating point operations per second) at speeds of 5.7 GHz while data transfer rates peak at 2.92 Terabits per second. Rambus has also unveiled new data transfer models that could easily go up to 1 Terabyte per second—about 20 times the intra-chip data transfer rates in the PlayStation 3.

How is this being implemented? In Intel's case, each core has a 5-port message passing router which is connected in a two dimensional mesh network with other cores that implements message passing—an efficient LAN on a chip; cores will communicate with each other better, and the router will ensure that the right data goes to the right chip. This architecture is much more scalable than present day multi-core chip interconnects in terms of speed and performance. On the Rambus side, we see new techniques like Full Differential Memory Architecture (which means in plain English, sending electrical signals across different wires at the same time, instead of across only one wire), 32x signalling (transferring data at 32 bits per I/O operation per clock cycle, compared to DDR's 2 bits per I/O operation per clock cycle) and FlexLink for command / address routing (using the same type of high speed signals which are in the data lines on the command / address line).

These technologies are going to be in the market by 2010, the companies say. In this context, we had talked about IBM's technology where multi-core chips use light rays to transfer data from one core to another, eliminating physical connections altogether. If and when the technology is perfected, such chips are likely to be first used in servers and high-end computers, especially for gaming and specialised tasks like video editing and scientific modelling that

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demand a lot of processing power. However, by 2015, your 10-year old kid's computer will have these beasts sitting inside it. And quad core would look so, well, last decade.

Smart Wires

Back to life-sized scales. Networks of the future will be worlds apart from today's relatively dumb ones that simply carry packets from one end to the other. With the explosion of P2P, high definition video streaming and ubiquitous computing—projects like Internet 0 are looking at a scenario where even light bulbs and toasters would have IP addresses and will be hooked into the Internet—networks as we understand them today may be overwhelmed by the load of data. Already, doom-sayers have been predicting how the “exaflood” of data is going to clog the pipes, and there have been quite a few apocalyptic predictions. Those on the other side of the fence are less sure, and many believe that Judgement Day, when networks will just give up, is not far off.

That said, networks of the future will look different, and will possibly work on different protocols. Such networks will not need full time chaperoning which is the job of today's system administrators. Called *adaptive networks*, they would provide a personalised experience for every user, with minimum maintenance. Adaptive networks would also seamlessly support different applications like VoIP, video conferencing, video surveillance, wireless applications, computing on demand or any other new application that users throw at them. Currently, most networks have a one-size-fits-all philosophy when it comes to applications, and when newer demands arise on the network, it starts to show its cracks. Security is also an important aspect, and future networks will be able to detect security attacks and take steps to fight back. Almost like what the immune system does for our body.

This scenario seems to be a bit too Utopian, but already companies have started work on plans that would make this a reality. ProCurve, the networking arm of HP has a neat solution that would make networks of the future “think” they way they're supposed to. Each port on the router and switch on the network will have individual silicon chips that would monitor the traffic flow in the network. These mini-processors would have software installed that could be optimised to run different applications like VoIP, video streaming, P2P file sharing, virus scanning, *et al.* Because the software is programmable, ports can easily switch data types from, say, VoIP to video in the event of heavy demand.

Such architecture would ensure that the available capacity is optimised for all these bandwidth-hungry applications. Another big draw for this decentralised architecture is that there will be no single choke point in the network, therefore ensuring that even in the case of a disaster, some part of the network would always be up and running. Contrast this with today's scenario, where networks are only as strong as the weakest link—hubs, routers, switches. For naysayers who foresee difficulties like keeping the network software patched and updated in absence of any centralisation, ProCurve has a solution—use a



A comparison between the data flow rates in DDR2 and TBI